Solutions to exercises in Lecture 4

1) We need to convert U_{ijk} where

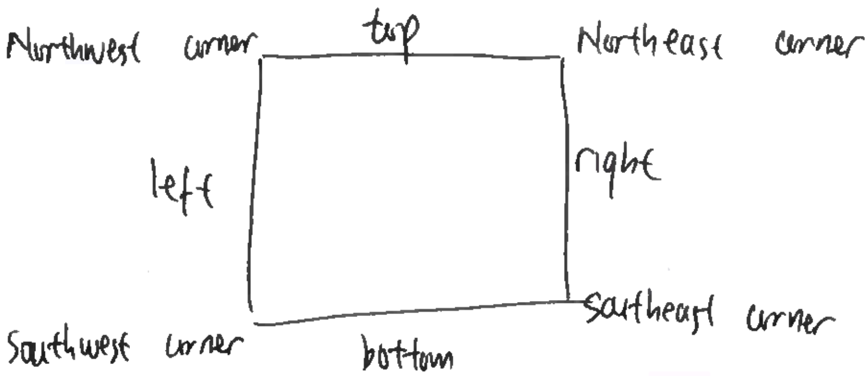
$$i = 0 \ldots (n_x - 1)$$
$$j = 0 \ldots (n_y - 1)$$
$$k = 0 \ldots (n_z - 1)$$

to a k-index scheme U_k.

This is not unique but one possibility is to write

$$k = k\, n_x n_y + j\, n_x + i$$

2) The modification depends on the location of the node.



Right interface / boundary

we use the central difference scheme to set:

$$\frac{\partial u}{\partial x} = \frac{u_{k+1} - u_{k-1}}{2\Delta x} = a$$

where a is a constant set by the Neumann BC

$$u_{k+1} = u_{k-1} + (2\Delta x)\, a$$

substituting this to the matrix equation for 2-D Laplace problem (see Eqs 4-6 in the lecture note):

$$u_{k-1} + (2\Delta x)\, a + u_{k-1} - 4 u_k + u_{k+n_x} + u_{k-n_x} = 0$$

$$2 u_{k-1} - 4 u_k + u_{k+n_x} + u_{k-n_x} = -(2\Delta x)\, a$$

so $A_{k,k+1} = 0$; $A_{k,k-1} = +2$; $A_{kk} = -4$; $A_{k,k+n_x} = 1$

$A_{k,k-n_x} = 1$; $b_k = -(2\Delta x)\, a$.

Using similar analysis, we can derive:

Left boundary: $A_{u,u+1} = 2$; $A_{u,u-1} = 0$; $A_{uu} = -4$

$A_{u,u+n_x} = 1$; $A_{u,u-n_x} = 1$; $b_u = (2\Delta X)\, a$

top boundary: $A_{u,u+1} = 1$; $A_{u,u-1} = 1$; $A_{uu} = -4$

$A_{u,u+n_x} = 0$; $A_{u,u-n_x} = 2$; $b_u = -(2\Delta Y)\, a$

bottom boundary: $A_{u,u+1} = 1$; $A_{u,u-1} = 1$; $A_{uu} = -4$

$A_{u,u+n_x} = 2$; $A_{u,u-n_x} = 0$; $b_u = (2\Delta Y)\, a$

For the corners, we need to apply the B.C both in x & y directions.
For example, for the north east corner, we apply:

$$\left.\begin{aligned} &\text{in x-direction}: && U_{u+1} = U_{u-1} + (2\Delta X)\, a \\ &\text{in y-direction}: && U_{u+n_x} = U_{u-n_x} + (2\Delta Y)\, a \end{aligned}\right\} \text{ assuming the same B.C applies}$$

Hence: $A_{u,u+1} = 0$; $A_{u,u-1} = 2$; $A_{uu} = -4$

$A_{u,u+n_x} = 0$; $A_{u,u-n_x} = 2$; $b_u = -2(\Delta X + \Delta Y)\, a$

Similarly,

south east: $A_{u,u+1} = 0$; $A_{u,u-1} = 2$; $A_{uu} = -4$

$A_{u,u+n_x} = 2$; $A_{u,u-n_x} = 0$; $b_u = 2(-\Delta X + \Delta Y)\, a$

north west: $A_{u,u+1} = 2$; $A_{u,u-1} = 0$; $A_{uu} = -4$

$A_{u,u+n_x} = 0$; $A_{u,u-n_x} = 2$; $b_u = 2(\Delta X - \Delta Y)\, a$

southwest: $A_{u,u+1} = 0$; $A_{u,u-1} = 2$; $A_{uu} = -4$

$A_{u,u+n_x} = 2$; $A_{u,u-n_x} = 0$; $b_u = 2(\Delta X + \Delta Y)\, a$.